Fuzzy Boundary-Aware Network for Hyperspectral Individual Tree Fine Recognition

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***Abstract*— Different tree species have different carbon storage and growth rates. Therefore, accurate segmentation and identification of individual trees can provide more detailed carbon storage data, which is the basis for accurately estimating forest carbon storage. However, individual tree segmentation and recognition in dense forest areas face challenges such as crown overlap, complex terrain, and species diversity. To address these challenges and improve recognition accuracy, this article proposes a fuzzy boundary-aware network (FBAN) for hyperspectral individual tree segmentation and recognition in dense forests. The proposed FBAN includes a boundary-aware module (BAM) that explores channel boundaries between trees and nontrees, spatial boundaries of trees, and spectral boundaries between different trees by integrating channel attention, spatial attention, and spectral attention. This enhances the separability of individual trees, especially those of the same species that are contiguous in dense forest areas. Additionally, an adaptive crown-aware module (ACAM) is constructed to adapt diverse-size crown features by coupling Transformer layers with dilated convolution layers. The experimental results on different hyperspectral datasets show that the proposed FBAN network outperforms existing methods in dense forest areas and different tree canopy areas, e.g., the AP on the SZU-South dataset is 4.5 points higher than that of Mask2former. It not only improves the accuracy of individual tree segmentation and recognition but also exhibits high generalization and robustness.**

***Index Terms*— Forest carbon storage, hyperspectral datasets, individual tree segmentation.**

I. Introduction

FORESTS, as crucial economic and environmental assets, play an irreplaceable role in carbon sequestration and fixation [1]. The country has bolstered forestry carbon sinks [2] through extensive measures, such as promoting land greening, safeguarding forest resources, enhancing forest management, and elevating forest quality. Research on the segmentation and identification of individual trees can accurately obtain the distribution and attribute information of individual trees, provide important support for research and management in various fields, and contribute to sustainable forest management and resource planning.

Individual tree species segmentation and identification [3] refer to segmenting the image of a forest or woodland area into individual trees while simultaneously identifying and classifying each tree to determine its species, thereby realizing the automated analysis of the identification and distribution of different tree species in the forest. Traditional tree species identification methods rely on manual field surveys, which are labor-intensive, costly, and time-consuming, making it impractical for large-scale mapping of extensive regions [4]. The rapid development of remote sensing technology has provided abundant technical means for forest tree species identification. The technology's macroscopic, real-time, and periodic characteristics facilitate the rapid, accurate, and efficient acquisition of large-scale forest resource information. Consequently, it is widely applied in the segmentation and identification of forest tree species.

Many researchers have successfully applied remote sensing technology to forest tree species identification and have achieved phased research results. Burt et al. [5] utilized the treeseg algorithm to automatically segment trees on point cloud data obtained from structurally simple open forests and complex tropical forests, achieving segmentation accuracies of 96% and 70%, respectively. Ferreira et al. [6] applied a fully convolutional neural network model to detect individual Amazon palm trees and classify tree species in unmanned aerial vehicle (UAV) RGB images. Chen et al. [7] subdivided the point clouds of different tree species into multiple voxels and then used the pointnet framework and height information to roughly segment the crown and refine the crown boundaries. Sun et al. [8] employed the point cloud segmentation (PCS) algorithm for individual tree segmentation in mature spruce forests. The aforementioned studies demonstrate that outstanding segmentation and recognition results can be obtained in forest data with a single tree species and relatively simple tree shapes.

However, there are challenges such as oversegmentation or undersegmentation in the segmentation and recognition of

Received 20 March 2025; revised 12 May 2025; accepted 16 June 2025. Date of publication 26 June 2025; date of current version 3 July 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 62271327; in part by the Project of Department of Education of Guangdong Province under Grant 2023KCXTD029; in part by Guangdong Basic and Applied Basic Research Foundation under Grant 2022A1515011290; in part by Shenzhen Science and Technology Program under Grant RCJC20221008092731042, Grant JCYJ20220818100206015, and Grant KQTD20200909113951005; in part by the Research Team Cultivation Program of Shenzhen University under Grant 2023JCT002; and in part by the General Program of Shenzhen Natural Science Foundation under Grant JCYJ20240813142308012. *(Corresponding author: Sen Jia.)*

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Digital Object Identifier 10.1109/TGRS.2025.3582412

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individual trees in more complex forest-type data. Dai et al. [9] initially performed rough segmentation of tree crowns using spatial geometric features. Then, they segmented undersegmented tree crowns using multispectral information. The segmentation results of this method are superior to those obtained using spatial features alone. Yin and Wang [10] found that when using UAV light detection and ranging (LiDAR) data to perform individual tree detection and delineation on mangroves, oversegmentation occurs due to the high cluster density and limited height difference between adjacent mangroves. Yang et al. [11] studied the impact of vegetation features such as tree density and canopy coverage on the individual tree segmentation methods in different forest types. The results showed that tree density and clustering are crucial vegetation features that affect tree detection accuracy, and this result is consistent with the conclusion of Vauhkonen et al. [12]. Combining the above research findings, existing individual tree segmentation methods can meet practical needs in forests with lower canopy closure. However, further research is still needed to address the segmentation of individual tree crowns in highly dense forest areas.

In densely forested areas, there are many technical challenges, such as higher species richness, greater stand density, complex canopy structure, confusing crown outlines, and overlap between crowns. In recent years, the rapid development of UAV remote sensing [13] platforms has provided new opportunities for individual tree segmentation and recognition. UAV hyperspectral data [14] have the characteristics of easy acquisition, low cost, and short cycle. Its images have a high spatial resolution, reaching submeter level, and can clearly capture the details and spatial features on the surface. Schiefer et al. [15] demonstrated that the high spatial features provided by high-resolution UAV imagery facilitate fine-scale tree species segmentation. Additionally, the spectral range of hyperspectral images (HSIs) can extend from visible light, near infrared, and even to mid-infrared. The number of bands can reach dozens to hundreds, forming a continuous and full spectral reflectance curve that can capture the subtle spectral features of ground objects. Although some progress has been made in the field of tree species segmentation using hyperspectral imagery, existing studies often focus on specific aspects. For example, Li et al. proposed an improved Mask R-CNN instance segmentation method based on UAV HSIs [16], aiming to identify individual diseased trees at different stages of pine wilt. Miyoshi et al. [17] proposed a deep learning method based on UAV HSIs to estimate the hyperspectral band combination that contributes most to the network architecture in the tree recognition task. Maschler et al. [18] used hyperspectral data combined with a mean shift segmentation algorithm to automatically extract tree crowns and applied random forest to classify 13 tree species in a temperate Austrian forest, showing that the classification accuracy of automatic segmentation was comparable to manual delineation. Tochon et al. [19] proposed an HSI segmentation method based on the binary partition tree algorithm, applied to canopy segmentation in tropical rainforests in Hawaii and Panama. Moreover, Long et al. [20] proposed a superpixel-guided scale pyramid graph network to achieve tree instance segmentation through node classification and edge prediction of the graph structure. However, existing studies mostly focus on the classification of single tree species or the segmentation of individual trees in sparse forest areas. Research specifically targeting the segmentation of individual tree crowns in dense forest environments is extremely rare. Therefore, segmenting and identifying individual tree crowns in these areas remains a research area that requires further exploration.

Previously, Ferreira et al. [21] found that visible light, near infrared, and short-wave infrared are the key bands for the classification and identification of forest tree species. Within these bands, the differences in the physical and chemical characteristics and water content of different vegetation can be reflected in the spectral reflectance. These differences provide important features for the segmentation and identification of forest tree species. The feature information of samples largely determines the accuracy of segmentation and recognition. In other words, the extraction and selection of texture features, geometric features, and spectral features is a key issue in individual tree segmentation and recognition.

Therefore, this article proposed a fuzzy boundary-aware network (FBAN) to fully extract the spatial–spectral joint features of trees from UAV hyperspectral imagery, effectively solving the problem of fuzzy boundaries in densely forested areas. This method constructs a boundary-aware module (BAM) to extract the channel boundary features of trees and nontrees, the spatial contour features of trees, and the differential spectral boundary features of trees. Especially when multiple trees of the same species overlap, the spectral weight of each tree is calculated through the spectral attention layer, which significantly highlights the spectral differences between different trees and extracts the spectral boundary features of individual trees. In addition, an adaptive crown-aware module (ACAM) is designed, which introduces Transformer layers and dilated convolution layers to adaptively fusion the overall structural features and local detail features of trees at various scales, enhancing the robustness and generalization ability of the method. Next, multiresolution features from our backbone are fed into a pixel decoder for progressive spatial recovery and high-level pixel embedding generation. Finally, these embeddings interact with object queries in a Transformer decoder to produce object embedding for individual tree mask prediction and classification. The main contributions of this article can be summarized as follows.

1) In dense forests, tree crowns often overlap, making it difficult to clearly distinguish the boundaries of individual trees. Therefore, the BAM module is designed to gradually extract the channel boundaries between trees and nontrees, the spatial contour boundaries of individual trees, and the different spectral boundaries between different trees through the channel attention layer, spatial attention layer, and spectral attention layer. In particular, when multiple crowns of the same tree species overlap, the spectral attention mechanism can further effectively segment and identify individual trees through measurable spectral difference information.
2) The size and shape of individual tree crowns vary significantly with tree species, growing environment,

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and age, presenting challenges for the precise segmentation and identification of individual trees. To overcome this challenge, the ACAM module is constructed, which parallelly couples Transformer layers and dilated convolution layers to capture multiscale features of different tree crowns. Subsequently, convolution operations dynamically adjust the fusion strategy of local and global features based on variations in the input features, enhancing the flexibility and adaptability of feature fusion.

3) An end-to-end network for individual tree segmentation and recognition based on HSIs is proposed, effectively eliminating cumbersome preprocessing and postprocessing steps. Experiments on diverse datasets verify that the model effectively segments and recognizes individual trees in dense and complex scenes, demonstrating strong robustness and generalization capabilities. This offers an efficient and reliable solution for tree species segmentation and recognition in HSIs.

The rest of this article is structured as follows. Section II provides a brief overview of related work. Section III presents a detailed explanation of the proposed method. Following that, Section IV systematically discusses the experimental settings and results. Finally, Section V presents the conclusions.

II. Related Work

A. Instance Segmentation

Instance segmentation identifies and segments each object instance in an image, distinguishing it from semantic segmentation by assigning unique identifiers to each instance. It is widely used in fields like autonomous driving, medical imaging, and object tracking. With deep learning advancements, many effective neural network-based methods have emerged, significantly improving performance in this task.

Mask region-based convolutional neural network (Mask R-CNN) [22] is a two-stage instance segmentation method proposed by He et al. [23]. It extends Faster R-CNN by adding a branch to generate precise binary masks for each detected instance, enabling fine-grained instance segmentation. Subsequently, the Cascade Mask R-CNN [24] further improves the performance by introducing a multistage cascade structure. The fundamental design includes an independent region proposal network for proposal generation, region of interest pooling for feature extraction, and separate classification and segmentation heads for their respective tasks. Hybrid task cascade (HTC) [25] is a multistage model for object detection and instance segmentation that combines Mask R-CNN and Cascade R-CNN. It uses a bounding box branch for mask output and adds a semantic segmentation branch to enhance performance. In 2019, the classic single-stage real-time instance segmentation method YOLACT [26] was proposed. The YOLACT model integrates the segmentation task into a single-stage network by performing parallel computation of object detection and instance segmentation, significantly improving processing speed. Building on this, the BlendMask [27] algorithm was further developed, introducing the Blender module to effectively combine instance-level global information with fine-grained low-level features, demonstrating excellent performance in processing complex scenes.

Edge imprecision is a major challenge in image segmentation. To address this, Kirillov et al. [28] proposed PointRend, which refines edges by predicting the Top N most uncertain points and using interpolation for others, significantly improving edge accuracy. Tsinghua University and other institutions later proposed RefineMask [29], which introduces the semantic head and mask head, combining a multistage refinement strategy to effectively enhance the model's boundary awareness.

Recent innovations leverage query for feature interaction and task unification, achieving breakthroughs in instance segmentation. Maskformer [30] uses Transformer cross-attention to unify semantic and instance segmentation, generating masks and class predictions via queries. Building on Maskformer, Mask2former [31] improves segmentation in complex scenes with multiscale feature fusion and enhanced cross-attention, offering a unified architecture for semantic, instance, and panoptic segmentation. The prototype-based efficient Maskformer model (PEM) [32] further optimizes the cross-attention mechanism and introduces a multiscale feature pyramid network (FPN), enhancing feature extraction and fusion capabilities. FastInst [33] is based on the meta-architecture of Mask2former, incorporating instance activation-guided queries, a dual-path update strategy, and ground truth mask-guided learning. This reduces the complexity of the pixel decoder and the number of Transformer decoder layers while delivering improved performance.

As instance segmentation methods advance, new technologies continue to emerge. However, individual tree segmentation and recognition remain in the early research stages, with most methods relying on the Mask R-CNN model. These studies indicate that the Mask R-CNN model [34] is an innovative approach for tree species recognition and crown width parameter extraction. For complex scenes such as mixed forests and urban forests, researchers have made improvements to Mask R-CNN. The proposed attention complementary network (ACNet) and edge detection R-CNN (ACE R-CNN) [35] provide a new solution for the task of tree species identification in high-density forests. In addition, to address band redundancy, some scholars have explored the impact of dimensionality reduction methods on instance segmentation networks, such as BlendMask and Mask R-CNN [36], for urban forest canopy delineation, demonstrating the stability of band selection methods. In plantation forest studies, Mask R-CNN not only enables the automatic detection of tree crowns but also extracts structural parameters, such as tree height, by integrating drone optical imagery and LiDAR data [37]. Moreover, some studies have adopted Faster R-CNN as the basic framework, integrating Swin Transformer as the backbone network for feature extraction, effectively capturing global information in complex scenes and enhancing tree detection accuracy [38]. In summary, while Mask R-CNN and its improvements show promise in tree species segmentation, research is still in its early stages. Future work should focus on more efficient feature extraction and other methods to enhance model robustness and generalization.

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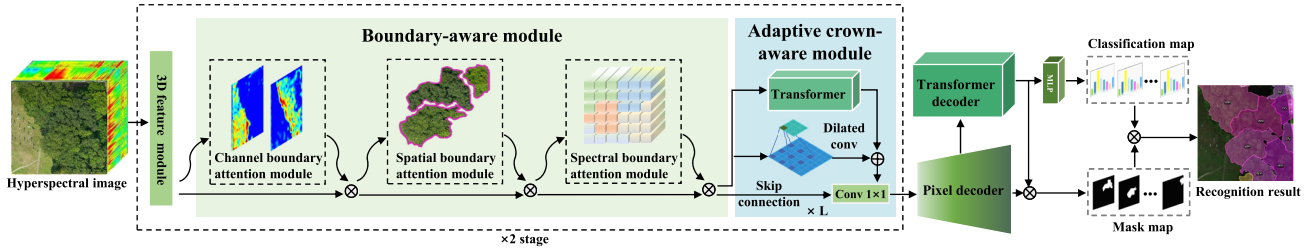


Fig. 1. Overall framework of the proposed FBAN for hyperspectral individual tree fine recognition in dense forest area, where ⊗ denotes the element-wise multiplication and ⊕ represents the element-wise addition. MLP stands for multilayer perceptron.

B. HSI Feature Extraction

HSIs contain rich spectral information and spatial information and have the characteristics of spectral continuity, image, and spectrum integration. It is widely applied in various fields such as precision agriculture, ecological environment monitoring, urban remote sensing, and military target detection. Hyperspectral feature extraction is a key task in HSI processing, which aims to make full use of the rich spectral–spatial information of hyperspectral data to improve the accuracy and efficiency of ground object segmentation and classification.

Traditional spectral feature extraction methods include principal component analysis (PCA) [39], independent component analysis (ICA) [40], local Fisher's discriminant analysis (LFDA) [41], and so on. However, the phenomenon of "different objects with the same spectrum" and "same objects with different spectra" widely exists in HSIs, which leads to certain limitations in simply extracting spectral information for image processing. Introducing spatial information into the feature extraction process can reduce the uncertainty of ground object recognition. The extended morphological profile (EMP) [42] extracts spatial structural information from HSIs through morphological operations. Apart from directly extracting spatial features, there are also approaches for extracting spectral–spatial fusion features from HSIs. For example, typical representatives include collaborative representation (CR) methods [43] and support vector machine (SVM) methods [44].

Traditional extraction of spectral–spatial features from HSIs mainly relies on manual operations, requiring professional experience and parameter settings. Moreover, it only processes the surface level, which to some extent limits the comprehensive representation and learning of features. Deep learning learns representative and discriminative features in data hierarchically and utilizes the superior information representation capabilities of deep structures to achieve automatic extraction and representation of features and eliminate dependence on manual feature extraction. The LeNet model [45] is a five-layer network and was the first successful application of convolutional neural networks to practical tasks. ResNet [46] is a classic deep learning model that adopts a residual network structure, effectively addressing the issues of gradient explosion and vanishing gradients caused by deepening the model. This allows the network to model more deeply, and hence, many subsequent models choose ResNet as the feature extraction network. In addition, attention-based feature extraction methods have rapidly evolved, emphasizing the identification and emphasis of the most informative and critical parts of the data rather than just focusing on extracting all information. Typical attention mechanisms include self-attention mechanisms, spatial attention mechanisms, and temporal attention mechanisms, with corresponding models being Transformer [47], convolutional block attention module (CBAM) [48], and long short-term memory (LSTM) [49], respectively. The feature extraction network is the backbone structure of the deep learning network. The quality of its design directly affects the performance of feature extraction, and its importance cannot be ignored.

III. Methodology

The dense forest area has rich species, high forest density, and complex canopy structure, resulting in confusing and unclear tree crown outlines. To address this challenge, this article proposes an FBAN-based segmentation and recognition method for dense individual trees in UAV HSI. This method can more accurately capture the spatial–spectral features of the tree crown, thereby achieving fine segmentation and recognition of individual trees. Next, we describe the overall process and some key designs of the model in detail. The framework of the proposed method is shown in Fig. 1.

A. 3-D Feature Module

Let $\mathbf{X} \in \mathbb{R}^{H\times W\times C}$ be hyperspectral data, where H, W, and C represent its height, width, and band number, respectively. First, it is input into the linear layer to obtain the pixel-level embedding $\mathbf{S}_{\text{pixel}}$

$$\mathbf{S}_{\text{pixel}} = \text{Conv}_{1\times1}(\mathbf{X}) \tag{1}$$

where $\text{Conv}_{1\times1}(\cdot)$ refers to the 2-D convolution function with a kernel size of 1×1. Meanwhile, it is fed into a 3-D feature module to model the 3-D structure of HSIs so as to jointly capture local spatial–spectral features. It is responsible for highlighting local spectral correlations, which is critical in HSI processing. The operation of the 3-D feature module is shown as follows:

$$\mathbf{F}_m = \text{RELU}(\text{BN}(\text{Conv}_{3\times3\times3}(\mathbf{X}))) \tag{2}$$

where $\text{Conv}_{3\times3\times3}(\cdot)$ refers to the 3-D convolution function with a kernel size of $3 \times 3 \times 3$. The 3-D convolutional module network consists mainly of two layers of convolutional neural networks, each layer containing four 3-D convolutional kernels. $BN(\cdot)$ represents the batch normalization function.

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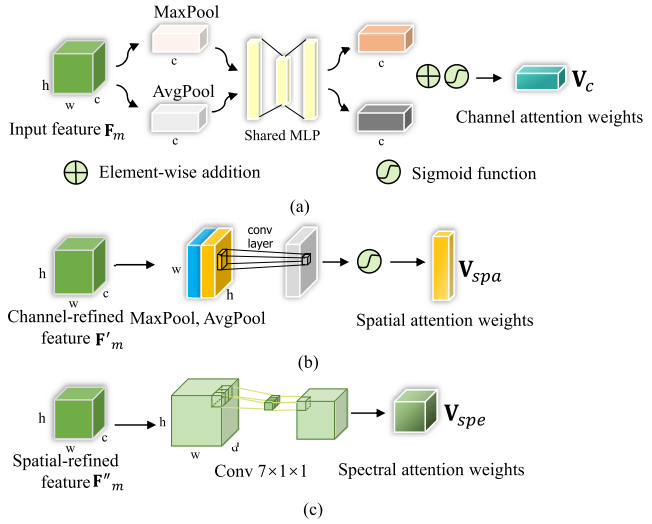


Fig. 2. Detailed structure of the BAM module. $\mathbf{F}_m$ is the input feature. $\mathbf{V}_c$, $\mathbf{V}_{\text{spa}}$, and $\mathbf{V}_{\text{spe}}$ are the channel attention weights calculated by the channel boundary attention module, the spatial attention weights calculated by the spatial boundary attention module, and the spectral boundary weights calculated by the spectral boundary, respectively. (a) Channel boundary attention module. (b) Spatial boundary attention module. (c) Spectral boundary attention module.

B. Boundary-Aware Module

After utilizing 3-D convolution to simultaneously process the spatial and spectral dimensions of hyperspectral data, the BAM is then employed to extract individual tree crown boundary features, capturing the fine spectral differences and complex spatial structures. Specifically, the BAM utilizes channel attention, spatial attention, and spectral attention to, respectively, extract the channels for trees and nontrees, the spatial shapes of trees, and the crucial spectral differences for each pixel of each tree, as illustrated in Fig. 2. This enables the extraction of channel boundaries, spatial boundaries, and spectral boundaries of individual trees from coarse to fine, thereby obtaining more accurate boundary contour information for trees in dense regions. The specific steps are described as follows.

First, the channel boundary attention module utilizes channel attention to automatically learn the importance of each channel based on the input data features, dynamically adjusting the weights of the channels. Through this approach, the module can selectively enhance channels that are relevant to tree morphology, texture, or other related features while suppressing irrelevant channels. Consequently, the module effectively guides the model's attention to channels relevant to tree features, thereby enhancing the model's ability and efficiency in extracting tree features. Specifically, the 3-D feature $\mathbf{F}_m$ undergoes global average pooling and max pooling to extract channel descriptors, which are processed through a shared MLP and summed to generate channel weights. These weights are then activated using the sigmoid function to produce the channel attention weights $\mathbf{V}_c$. Finally, $\mathbf{V}_c$ is multiplied channel by channel with the feature $\mathbf{F}_m$ to obtain the output feature $\mathbf{F'}_m$

$$\mathbf{F'}_m = \mathbf{V}_c \otimes \mathbf{F}_m \tag{3}$$

$$\mathbf{V}_c = \sigma\left(\mathbf{V}_c^1 + \mathbf{V}_c^2\right) \tag{4}$$

$$\mathbf{V}_c^1 = \text{MLP}(\text{AvgPool}(\mathbf{F}_m)) \tag{5}$$

$$\mathbf{V}_c^2 = \text{MLP}(\text{MaxPool}(\mathbf{F}_m)) \tag{6}$$

where $\sigma(\cdot)$ represents the sigmoid function and $+$ means the element-wise addition; MLP$(\cdot)$ represents the shared multilayer perceptron module; and AvgPool$(\cdot)$ and MaxPool$(\cdot)$ represent average-pooling operation and max-pooling operation, respectively. Average pooling can resist the interference of noise, while maximum pooling can highlight the significant features.

Moreover, different tree species have differences in crown shape, leaf shape distribution, and trunk thickness. Therefore, the spatial boundary attention module further utilizes spatial attention mechanisms to focus more on positions relevant to the spatial morphology of trees. It adaptively adjusts the weights of different spatial positions, enabling accurate extraction of spatial boundary features for different tree species. Consequently, the feature $\mathbf{F'}_m$ is subjected to channel-wise average pooling and max pooling to generate spatial feature maps, which are stacked and processed with a 7×7 convolution to produce spatial attention weights $\mathbf{V}_{\text{spa}}$. After normalization, these weights are applied element-wise to $\mathbf{F'}_m$, resulting in the output $\mathbf{F''}_m$

$$\mathbf{F''}_m = \mathbf{V}_{\text{spa}} \otimes \mathbf{F'}_m \tag{7}$$

$$\mathbf{V}_{\text{spa}} = \sigma\left(\text{Conv}_{7\times7}\left(\mathbf{V}_{\text{spa}}^1\right)\right) \tag{8}$$

$$\mathbf{V}_{\text{spa}}^1 = \left[\text{AvgPool}\left(\mathbf{F'}_m\right); \text{MaxPool}\left(\mathbf{F'}_m\right)\right] \tag{9}$$

where $\text{Conv}_{7\times7}(\cdot)$ represents the convolution operation with a filter size of 7×7.

Additionally, it is generally known that different tree species exhibit unique spectral characteristics due to variations in their biochemical composition, leaf structure, and physiological processes. However, it is worth noting that even within the same tree species, different individuals may have subtle spectral differences. The reason is that asymmetric competition among plants indicates an imbalance in competition between different individuals of the same tree species. Specifically, when certain individuals can acquire resources more efficiently (such as water and light), it may lead to resource limitations for other individuals, thereby causing changes in their spectral curves. Therefore, in dense regions, especially when dealing with contiguous instances of the same tree species, extracting significant spectral difference information between different trees becomes particularly important. The spectral boundary attention module receives the input spatial optimization feature $\mathbf{F''}_m$ and applies a convolution operation with a $7\times1\times1$ kernel on the reflectance values or radiance values across a series of spectral bands for each pixel, generating the spectral weight $\mathbf{V}_{\text{spe}}$ for each pixel. It assigns distinct spectral weights to different tree species, enabling the formation of spectral boundaries between different trees that are contiguous or overlapping within dense regions. Therefore, the spectral boundary weight for each individual tree can be calculated as follows:

$$\mathbf{F'''}_m = \mathbf{V}_{\text{spe}} \otimes \mathbf{F''}_m \tag{10}$$

$$\mathbf{V}_{\text{spe}} = \sigma\left(\text{Conv}_{7\times1\times1}\left(\mathbf{F''}_m\right)\right) \tag{11}$$

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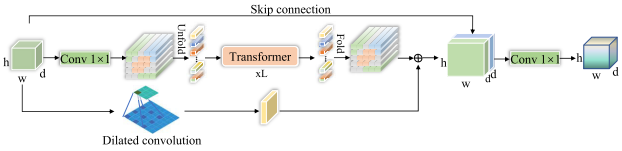


Fig. 3. Detailed structure of the ACAM module, where $\oplus$ represents the element-wise addition.

where $\mathbf{F}'''_m$ represents the features processed by the spectral attention module. Finally, the features processed by the entire BAM are represented as $\widehat{\mathbf{F}}_m$, and the entire process can be summarized as follows:

$$\begin{aligned}\widehat{\mathbf{F}}_m &= \mathbf{F}'''_m \\ &= \mathbf{V}_{\text{spe}} \otimes \mathbf{F}''_m \\ &= \mathbf{V}_{\text{spe}} \otimes \left(\mathbf{V}_{\text{spa}} \otimes \mathbf{F}'_m\right) \\ &= \mathbf{V}_{\text{spe}} \otimes \left(\mathbf{V}_{\text{spa}} \otimes (\mathbf{V}_c \otimes \mathbf{F}_m)\right)\end{aligned} \tag{12}$$

where $\widehat{\mathbf{F}}_m$ is the feature after boundary awareness processing by BAM and $\otimes$ denotes the element-wise multiplication.

Meanwhile, performing dot product computation between the pixel-level embedding $\mathbf{S}_{\text{pixel}}$ and the attention map $\mathbf{V}_{\text{spe}}$ from BAM achieves embedding weighting, thereby obtaining the weighted semantic feature $\tilde{S}_{\text{pixel}}$. Its formula is

$$\tilde{S}_{\text{pixel}} = \mathbf{S}_{\text{pixel}} \otimes \mathbf{V}_{\text{spe}}. \tag{13}$$

Thus, by embedding spectral information and integrating semantic information, the expression of tree species characteristics is enhanced.

C. Adaptive Crown-Aware Module

Further research found that the crown size and shape of trees exhibit various changes and adaptations during their growth due to multiple factors, including growth stages, survival competition, and human influences. Therefore, this module couples Transformer and dilated convolution to extract global overall information and local detail information of different crowns, effectively solving the problem of tree segmentation at different crown scales, as shown in Fig. 3. Specifically, the embedding feature is first reencoded using a 2-D convolutional layer with a kernel size of 1×1

$$\Gamma = \text{Conv}_{1\times1}\left(\text{Concat}\left(\widehat{\mathbf{F}}_m, \tilde{S}_{\text{pixel}}\right)\right) \tag{14}$$

where Concat$(\cdot)$ is a concatenated operation along the channel dimension.

Following that, the features are flattened along the spatial dimension to create the input sequence of the Transformer encoder. The encoder models the global relationship of different trees and regains a global feature with unchanged relative position

$$\widehat{\Gamma}_{\text{TR}} = \text{Fold}(\text{TR}(\text{Unfold}(\Gamma))) \tag{15}$$

where TR$(\cdot)$ represents the standard Transformer encoder module without positional encoding and Unfold$(\cdot)$ and Fold$(\cdot)$ denote spatial unfolding and spatial folding operations for matrices, respectively.

Since there is no pooling operation in the dilated convolution, the information loss problem caused by pooling can be avoided. The dilated convolution can expand the receptive field of the convolution kernel without increasing the number of parameters, thereby capturing the features of small objects. This is especially important for the segmentation and recognition of small tree crowns. Therefore, the dilated convolution [50] is used to extract the features of small tree crowns

$$\widehat{\Gamma}_{DI} = \text{DilatedConv}_{3\times3}(\Gamma) \tag{16}$$

where DilatedConv$(\cdot)$ represents the dilated convolution function.

Then, the module combines and integrates the output features from both branches, incorporates input information to the module through skip connections, and utilizes convolutional adaptive fusion to generate the module's output

$$\widehat{\Gamma} = \text{Conv}_{1\times1}\left(\text{Concat}\left(\widehat{\Gamma}_{TR} + \widehat{\Gamma}_{DI}\right), \Gamma\right) \tag{17}$$

where Concat$(\cdot)$ denotes the concatenation operation by channel dimension and $+$ represents the element-wise addition. This module adaptively fuses the local and global features of the tree through convolution according to the original input and dynamically adjusts the weights between feature layers. Thus, this module can effectively handle tree crown structures of different sizes and shapes, enhance the flexibility and accuracy of the model, and significantly improve the generalization performance of the model under diverse environmental conditions.

D. Mask Attention Segmentation Module

The low-resolution features extracted from the image by the above backbone feature extractor are input into the pixel decoder. Gradually, they are upsampled to generate a feature pyramid composed of high-resolution features with resolutions of 1/16, 1/8, and 1/4 of the original image. The high-resolution features provide more detailed information, enabling the model to more easily distinguish and recognize small-scale objects. The different scale features of the multiscale features are then fed to different Transformer decoder layers. For each resolution, a sinusoidal positional embedding $\varepsilon_{\text{sin}} \in \mathbb{R}^{H_l W_l \times C}$ and a learnable scale embedding $\varepsilon_{lea} \in \mathbb{R}^{1\times C}$ are added. Therefore, the first three layers obtain feature maps with resolutions $H_1 = H/16, H_2 = H/8$, and $H_3 = H/4$, and $W_1 = W/16, W_2 = W/8$, and $W_3 = W/4$. Repeating this three-layer Transformer decoder L times, the final Transformer decoder therefore has $3L$ layers. The Transformer decoder utilizes image features to process instance queries, and the final binary mask prediction is decoded from the pixel-wise embedding via the instance query. Key components of the Transformer decoder include a masked attention operator, which extracts local features by constraining cross-attention within the foreground region of the prediction mask for each query, rather than focusing on the entire feature map. Masked attention adjusts the attention matrix by adding mask values. The formula for masked attention can be expressed as follows:

$$\mathbf{A}_l = \text{softmax}\left(\mathbf{P}_{l-1} + \mathbf{Q}_l \mathbf{K}_l^T\right)\mathbf{V}_l + \mathbf{A}_{l-1} \tag{18}$$

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$$\mathbf{P}_{l-1}(x,y)=\begin{cases}0, & if\ \mathbf{P}_{l-1}(x,y)=1\\ -\infty, & \text{otherwise}\end{cases} \tag{19}$$

where l denotes the layer number and $\mathbf{A}_l \in \mathbb{R}^{M\times C}$ represents the M features each of C dimensions for the lth layer. $\mathbf{Q}_l$, $\mathbf{K}_l$, and $\mathbf{V}_l$, respectively, represent the query, key, and value information in the lth layer. $\mathbf{P}_{l-1} \in \{0,1\}^{M\times H_l W_l}$ is the binary output of the adjusted mask prediction from the previous $l-1$ Transformer decoder layers, with dimensions $M \times H_l W_l$. It is adjusted to have the same resolution as $\mathbf{K}_l$. $\mathbf{P}_0$ is the binary mask prediction obtained from $\mathbf{A}_0$, i.e., before feeding the query features into the Transformer decoder.

E. Loss Function

To measure the similarity between the masks generated by the model and the ground truth masks, binary cross-entropy and Dice loss functions are used to fit the mask regions of individual trees. This approach provides a comprehensive evaluation of mask accuracy and completeness, thereby improving the model's performance in individual tree segmentation tasks. Its mask loss function is

$$L_{\text{mask}} = \lambda_{\text{ce}} L_{\text{ce}} + \lambda_{\text{dice}} L_{\text{dice}}. \tag{20}$$

Next, the class cross-entropy loss function is used to fit the classes of individual trees. This loss function measures the difference between the model's predictions for tree classes and the ground truth, helping the model to better learn tree class information. The class loss function can be represented as $\lambda_{\text{cls}} L_{\text{cls}}$.

The final loss function combines the class loss function and the mask loss function, comprehensively considering the model's performance in tree classification and segmentation tasks. It can be expressed as

$$L = L_{\text{mask}} + \lambda_{\text{cls}} L_{\text{cls}} \tag{21}$$

where $\lambda_{\text{ce}} = 5.0$, $\lambda_{\text{dice}} = 5.0$, and $\lambda_{\text{cls}} = 2.0$.

IV. Experiments

A. Dataset

1) SZU-North Dataset: The SZU-North dataset was captured in 2022 at the north campus of Shenzhen University using a UAV equipped with a Specim FX10 hyperspectral camera. Specim FX10 is a high-speed line-scanning hyperspectral camera, capturing imagery within the wavelength range of 400–1000 nm. The obtained data have a total of 112 continuous bands after preprocessing, with a spatial resolution of 0.1 m per pixel and a spectral resolution of 5.5 nm. In this experiment, 256 images were cropped from the captured HSIs, with each image having dimensions of 512×512 pixels. The dataset was annotated based on field surveys that recorded the species and specific locations of the trees, containing a total of 5103 tree instances across 30 different tree species. Table I displays all tree species categories within the dataset along with their corresponding counts of tree instances. There are many types and numbers of trees in the area, and their spatial distribution is dense, with canopies often staggered, intertwined, or overlapping. Consequently, the scenes captured in the images exhibit rich details and variations. The specific scene is illustrated in Fig. 4.

TABLE I
Tree Species on the SZU-North Dataset and the Number of Training Masks and Test Masks for Each Tree Species

Class	Name	Code	Training mask	Test mask
1	Albizia falcataria	AJLR	146	56
2	Ficus altissima Blume	BQC	126	32
3	Ficus microcarpa	DG	230	90
4	Spathodea campanulata	DWYZ	79	46
5	Roystonea regia	DY	330	116
6	Delonix regia	DYR	74	26
7	Mangifera indica	FHM	302	86
8	Litchi chinensis	FLM	907	383
9	Leucaena leucocephala	HG	15	4
10	Eucalyptus citriodora Hook.f.	HYM	23	12
11	Ficus religosa	JBL	14	3
12	Acacia auriculiformis	JDH	174	44
13	Dracontomelon duperreanum	LY	201	82
14	Terminalia arjuna	LZ	69	13
15	Dimocarpus longan	MG	107	32
16	Livistona chinensis	MM	161	34
17	Bauhinia purpurea L	MMH	79	14
18	Plumeria rubra	NMA	13	10
19	Melaleuca leucadendron Linn.	NYS	85	9
20	Casuarina equisetifolia	NYY	57	19
21	Cinnamomum camphora	PK	89	76
22	Araucaria cunninghamii	PT	102	13
23	Ficus elastica	RMZ	10	2
24	Archontophoenix alexandrae	THXM	51	4
25	Swietenia mahagoni	WM	35	22
26	Ficus virens Aiton	XS	75	9
27	Kigelia africana	XYR	11	2
28	Lagerstroemia indica L.	YMM	58	25
29	Tabebuia chrysantha	YTJ	123	40
30	unknown	unknown	31	21
Total	Mask		3777	1325
	Image		192	64

TABLE II
Tree Species on the SZU-South Dataset and the Number of Training Masks and Test Masks for Each Tree Species

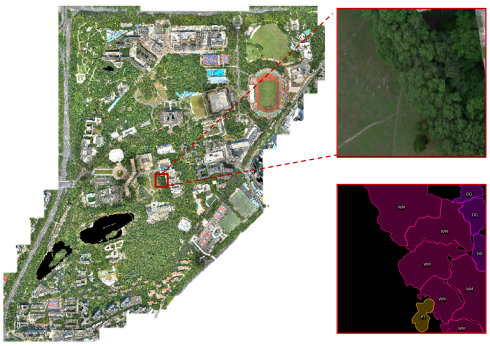
Class	Name	Code	Training mask	Test mask
1	Delonix regia	FHM	38	17
2	Artocarpus parvus	GM	76	10
3	Wodyetia bifurcata	HWYZ	159	22
4	Plumeria rubra	JDH	100	3
5	Phoenix sylvestris Roxb	LCK	102	31
6	Koelreuteria paniculata	LM	16	15
7	Choerospondias axillaris	NSZ	43	98
8	Celtis sinensis	PS	45	16
9	Senna siamea	TDM	184	105
10	Leucaena leucocephala	YHH	478	115
11	Cinnamomum camphora	ZS	357	81
12	Lagerstroemia indica	ZW	36	1
13	unknown	unknown	25	9
Total	Mask		1659	523
	Images		75	22

2) SZU-South Dataset: This dataset was captured in 2022 using a UAV equipped with a Specim FX10 hyperspectral camera over the south campus of Shenzhen University. Therefore, this dataset also contains 112 spectral bands, with a spatial resolution of 0.1 m per pixel and a spectral resolution of 5.5 nm. Based on field surveys, the dataset includes 2182 tree instances of 13 different tree species. Table II shows all tree species and their corresponding tree numbers. The trees in this dataset vary in size and canopy foliage density, as shown in Fig. 5. The second dataset for this experiment consists of 97 images cropped from the acquired images, with each image having dimensions of 532×532 pixels.

B. Experimental Setup

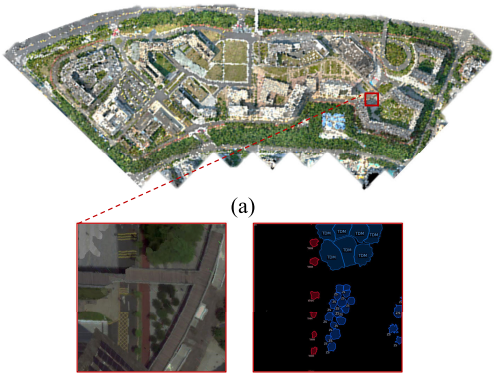
The main factors affecting model performance include hyperparameter adjustment, regularization technology,

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(a) (b)

Fig. 4. (a) North campus of Shenzhen University and (b) HSI image of the cropped scene and its corresponding ground truth.



(a)

(b)

Fig. 5. (a) South campus of Shenzhen University and (b) HSI image of the cropped scene and its corresponding ground truth.

optimizer selection, and so on. In the proposed FBAN individual tree segmentation model, the following factors are primarily considered.

1) Data Split: In the SZU-North dataset, 192 images are randomly selected, which contain 3778 annotated masks for training; the remaining 64 images contain 1325 masks for testing. The specific data of each class are shown in Table I. Similarly, in the SZU-South dataset, 75 images are randomly selected, containing a total of 1659 masks for training, and the remaining 22 images contained 523 masks for testing. The specific data of each class are shown in Table II. It is worth noting that the images in the training and test sets do not overlap and are ensured to cover all tree species. After partitioning the training set and test set, data augmentation is performed on the training set during the training phase. HSIs are processed using random flipping up and down, random left and right flipping, random proportional scaling (the scaling factor is about 0.1–2), and random cropping, resulting in final images of size 512×512. Additionally, mean and standard deviation normalization is applied to the HSIs. During the testing phase, the images are resized to 512×512.

2) Model Hyperparameter: In the model parameter settings, the channel dimension of the first-stage backbone feature extractor is set to 64, while the channel dimension of the second-stage backbone feature extractor is set to 128. In the pixel decoder, the low-resolution features are gradually upsampled to generate a feature pyramid consisting of high-resolution features with spatial resolutions of 1/4, 1/8, and 1/16. The Transformer decoder follows the same settings as mask2former.

3) Optimization: In the training process, the adamw optimizer is used with an initial learning rate set to 0.0001 and a weight decay set to 0.05. The learning rate is dynamically adjusted using warmup PolyLR. To prevent the gradient explosion problem, gradient clipping is used to constrain the L_2 norm of the gradients to not exceed 0.01. The model undergoes a total of 100 000 training steps, with a batch size set to 8 (i.e., eight images are used to train the model in each step).

TABLE III

ABLATION EXPERIMENT RESULTS OF FBAN UNDER DIFFERENT FEATURE EXTRACTION MODULES IN THE SZU-NORTH DATASET

Module	AP	AP_{50}	AP_{75}	AP_s	AP_m	Precision	Recall	F1 score
w/o_spectral_boundary	40.26	60.07	47.77	16.80	46.85	81.21	61.17	68.31
w/o_spatial_boundary	38.89	61.46	42.97	20.72	43.33	81.52	64.21	70.07
w/o_channel_boundary	41.48	61.45	47.40	19.15	47.41	83.39	62.78	70.88
w/o_spatial_spectral_ boundary	36.96	58.07	42.30	15.58	42.74	81.90	58.94	67.42
w/o_channel_spatial_spectral_ boundary	35.23	55.69	41.75	14.91	40.73	81.54	56.83	65.59
w/o_Transformer	42.35	62.03	49.46	20.54	**48.38**	81.93	63.38	70.22
w/o_dilated_convolution	41.06	58.27	50.21	20.56	44.79	79.28	59.21	66.82
w/o_skip_connection	38.94	58.83	45.84	16.28	45.16	81.72	60.05	67.82
FBAN	**44.08**	**65.29**	**50.49**	**22.01**	47.98	**87.20**	**65.60**	**73.47**

TABLE IV

ABLATION EXPERIMENT RESULTS OF FBAN UNDER DIFFERENT FEATURE EXTRACTION MODULES IN THE SZU-SOUTH DATASET

Module	AP	AP_{50}	AP_{75}	AP_s	AP_m	Precision	Recall	F1 score
w/o_spectral_boundary	50.35	76.11	53.04	32.70	68.63	88.65	77.29	81.49
w/o_spatial_boundary	48.24	74.95	48.62	28.23	69.28	**91.33**	76.90	82.65
w/o_channel_boundary	52.51	73.95	57.56	**40.72**	68.13	89.12	76.24	81.32
w/o_spatial_spectral_ boundary	47.10	75.91	48.64	27.66	67.93	85.74	73.19	78.15
w/o_channel_spatial_spectral_ boundary	44.15	71.82	43.57	29.49	61.39	82.62	69.26	74.59
w/o_Transformer	52.68	76.52	59.79	35.50	**73.38**	85.05	74.16	78.46
w/o_dilated_convolution	49.62	73.98	57.28	32.71	68.11	85.45	71.04	75.84
w/o_skip_connection	52.14	75.96	56.91	32.10	72.52	84.06	71.57	76.51
FBAN	**54.32**	**78.36**	**62.69**	39.59	72.03	89.28	**79.32**	**83.26**

C. Ablation Analysis

To verify the effectiveness and necessity of each module, ablation experiments are designed to identify and quantify their contributions to the model's overall performance.

It is evident from the second row in Tables III and IV that when the spectral boundary attention module is absent, the index of individual tree species segmentation is lower compared to the proposed FBAN framework. For example, in the SZU-North dataset, its AP and precision are 3.82% and 5.99% lower than FBAN, respectively. The reason is that the SZU-North dataset contains a diverse range of tree species, with trees densely packed and overlapping. Therefore, it is necessary to learn the subtle spectral differences between different trees to accurately segment the edge contours between different trees in complex and overlapping areas. From the second row in Tables III and IV, it can be seen that the $F1$-score in the SZU-North dataset decreases by 5.16%, while the $F1$-score of the SZU-South dataset is only reduced by 1.77%. This indicates that the performance drop in the FBAN framework is more significant in the SZU-North dataset when the spectral boundary attention module is removed. This is because the tree conditions in the SZU-North dataset are indeed more complex, with higher levels of tree overlap and canopy density compared to the SZU-South dataset. Consequently, the absence of this

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module hinders the effective extraction of spectral boundary information, making the performance degradation more pronounced.

It is particularly noteworthy that in the third row of the table, due to the absence of the spatial boundary attention module, the AP in Tables III and IV is only 38.89% and 48.24%, respectively. This indicates that spatial structural information indeed plays an important role, as different tree species have distinct canopy structures. Additionally, upon analysis of the fifth row in Table IV, it can be found that when both the spatial boundary attention module and the spectral boundary attention module are removed, leaving only the channel boundary attention module in the BAM, AP_{75} is only 48.64%, whereas the proposed FBAN framework achieves 62.69%. By comparing the first six rows of Tables III and IV, it can be observed that as each of the channel boundary attention module, spatial boundary attention module, and spectral boundary attention module is gradually removed, the segmentation performance metrics gradually decrease. It is more worth mentioning that when the BAM is removed from the FBAN framework, AP_{50} drops from 65.29% to 55.69%, a significant decrease of 9.6%.

By observing the seventh row in Table III, it can be seen that when the Transformer in the adaptive size-aware module is removed, precision drops from 87.20% to 81.93%, a decrease of 5.27%. The reason is that the tree crowns in the SZU-North dataset are generally larger, and the Transformer's ability to capture long-distance dependencies makes it excellent at extracting features from large-sized tree crowns and can effectively integrate the global information of tree species. However, its AP_m is slightly higher than that of other modules, indicating that dilated convolution or skip connections play a role here. Nevertheless, as seen in the ninth row of Table IV, the removal of skip connection here does not impact the segmentation performance. This indicates that dilated convolution plays a significant role in extracting medium-scale tree crown features due to its ability to preserve image details while expanding the receptive field and capturing more contextual information.

Since the datasets used in the experiments contain a relatively higher number of medium-scale tree crowns, AP_m in this case is slightly higher than that of other modules. The eighth row in Tables III and IV also proves this conclusion. Without dilated convolution, its AP_m is 3.19% and 3.92% lower than that of FBAN. In the ninth row of Table IV, when the skip connection is removed, the recall of the FBAN framework immediately drops from 79.32% to 71.57%, a decrease of 7.75%. The factors that contribute to this situation include the skip connection in the proposed FBAN framework, which combines low-level fine-grained features with high-level semantic features, thereby capturing the subtle characteristics and structural information of tree species more effectively.

D. Comparative Experiment

To validate the effectiveness of the proposed FBAN model, comparative experiments are conducted on two different types of tree species datasets. The comparison methods used in the experiments include Mask R-CNN [22], Cascade Mask R-CNN [24], Condinst [51], SOLOv2 [52], BoxInst [53], Transfiner [54], Maskformer [30], and Mask2former [31]. To ensure fairness, the parameters for all comparative methods are strictly set according to their original papers.

TABLE V
SEGMENTATION AND RECOGNITION RESULTS OF SEVEN DIFFERENT COMPARISON METHODS ON SZU-NORTH HYPERSPECTRAL DATASET

Method	AP	AP_{50}	AP_{75}	AP_s	AP_m	Confidence
Mask R-CNN	28.29	46.67	29.38	13.60	32.05	91.36
Cascade Mask R-CNN	29.50	46.79	32.69	14.36	31.50	91.00
Condinst	30.25	48.10	30.68	14.73	33.97	79.98
Solov2	19.36	28.74	20.28	11.01	21.54	84.70
BoxInst	9.05	35.23	1.58	6.12	11.01	87.09
Transfiner	30.86	49.20	32.80	15.52	34.96	95.05
Maskformer	17.77	26.02	19.50	9.07	19.84	87.67
Mask2former	40.57	61.24	46.08	20.03	44.70	94.68
FBAN	**44.08**	**65.29**	**50.49**	**22.01**	**47.98**	**95.57**

1) Experimental Results on SZU-North Dataset: Tables V and VI show the overall segmentation and recognition results of the proposed FBAN and other comparison methods on the SZU-North dataset, respectively. Figs. 6 and 7, respectively, show the ground truth segmentation labels and the visualized segmentation and recognition results predicted by the comparative models for some dense areas and areas of different crown sizes in the SZU-North dataset. As can be seen from Table V, BoxInst has the lowest AP value among all methods, especially for the segmentation of small object trees, which is only 6.12%. The average recall in Table VI is only 28.86%, indicating that the BoxInst method can only recognize less than one-third of the actual positive samples. The average $F1$-score is 36.82%, and this relatively low value further indicates that the BoxInst method has not achieved a good balance between precision and recall. In some classes, BoxInst also shows relatively good performance. For example, in the Litchi chinensis category, the accuracy reached 66.67%. However, its performance is poor in many other classes, such as *Roystonea regia* and *Kigelia africana*, where metrics like precision and recall are very low, with some even having values of 0. The main reason is that BoxInst primarily relies on coarse-grained bounding box information for boundary refinement, whereas individual tree segmentation requires fine-grained feature capture, particularly in areas with complex edges and multiple overlaps. As shown in Fig. 6, the segmentation boundary does not align well with the crown boundary, and there are instances of both missed and mis-segmentation.

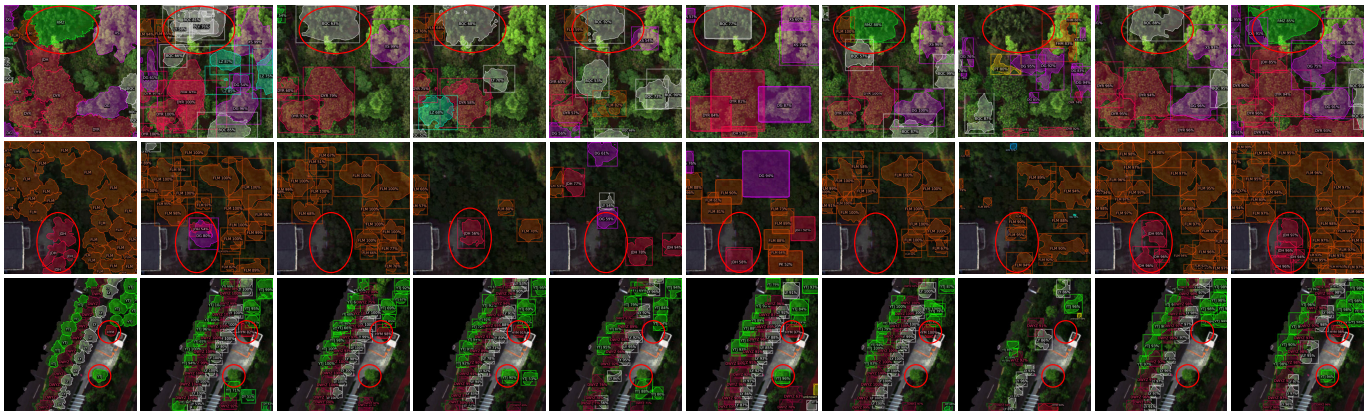
Second, the segmentation performance of Maskformer is also poor, with an AP value of only 17.77%, which is 26.31% lower than FBAN. In particular, its segmentation performance on small objects is unsatisfactory, with AP_s of only 9.07%. This can be clearly seen from the visualized segmentation and recognition results in the red-circled area in the second row of Fig. 7, where small-sized *Elaeocarpus decipiens* Hemsl. is missed. Perhaps the primary factor is that the self-attention mechanism considers information from all positions when processing each position, resulting in a global receptive field. Although this is beneficial for capturing global dependencies, it may ignore fine-grained local information to some

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TABLE VI

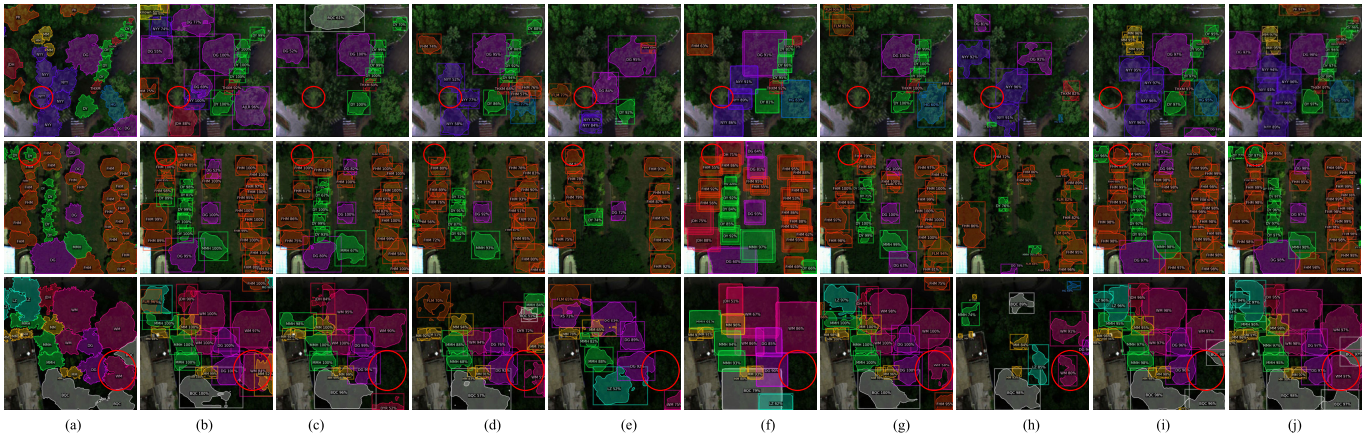
SEGMENTATION AND RECOGNITION RESULTS OF DIFFERENT CLASSES BY DIFFERENT COMPARISON METHODS ON SZU-NORTH HYPERSPECTRAL DATASET (BOLD NUMBERS INDICATE THE BEST CLASSIFICATION PERFORMANCE)

Class names	Precision									Recall									F1-score								
	Mask R-CNN	Cascade Mask R-CNN	Condinst	Solov2	BoxInst	Trans-finer	Mask-former	Mask2-former	FBAN	Mask R-CNN	Cascade Mask R-CNN	Condinst	Solov2	BoxInst	Trans-finer	Mask-former	Mask2-former	FBAN	Mask R-CNN	Cascade Mask R-CNN	Condinst	Solov2	BoxInst	Trans-finer	Mask-former	Mask2-former	FBAN
Litchi chinensis	61.54	80.00	72.73	50.00	66.67	65.00	20.83	90.91	**93.48**	57.14	42.86	57.14	35.71	39.29	46.43	17.86	71.43	**76.79**	59.26	55.81	64.00	41.67	49.44	54.17	19.23	80.00	**84.31**
Araucaria cunninghamii	32.35	62.50	42.86	27.78	**76.92**	47.83	36.36	57.14	66.67	34.38	31.25	28.13	15.63	31.25	34.38	25.00	**50.00**	43.75	33.33	41.67	33.96	20.00	44.44	40.00	29.63	**53.33**	52.83
Ficus microcarpa	64.71	77.27	75.56	57.78	61.22	75.00	34.33	**77.94**	75.76	48.89	37.78	37.78	28.89	33.33	46.67	25.56	**58.89**	55.56	55.70	50.75	50.37	38.52	43.17	57.53	29.30	**67.09**	64.10
Elaeocarpus decipiens Hemsl.	91.11	95.45	93.48	79.07	93.02	93.18	76.67	**100.00**	93.02	89.13	91.30	**93.48**	73.91	86.96	89.13	50.00	89.13	86.96	90.11	93.33	93.48	76.40	89.89	91.11	60.53	**94.25**	89.89
Melaleuca leucadendron Linn.	75.95	82.61	83.10	80.00	46.38	**92.54**	75.00	82.72	84.52	51.72	49.14	50.86	37.93	27.59	53.45	25.86	57.76	**61.21**	61.54	61.62	63.10	51.46	34.59	67.76	38.46	68.02	**71.00**
Ficus altissima Blume	62.50	64.71	69.23	77.78	62.50	78.57	31.58	**86.36**	84.21	57.69	42.31	34.62	26.92	38.46	42.31	23.08	**73.08**	61.54	60.00	51.16	46.15	40.00	47.62	55.00	26.67	**79.17**	71.11
Bauhinia purpurea L.	62.11	80.28	79.45	65.28	38.10	74.03	67.24	**93.67**	87.50	68.60	66.28	67.44	54.65	27.91	66.28	45.35	86.05	**89.53**	65.19	72.61	72.96	59.49	32.21	69.94	54.17	**89.70**	88.51
Acacia auriculiformis	45.45	63.10	50.78	34.94	32.87	56.30	17.77	68.00	**71.43**	32.64	30.81	34.20	22.72	18.54	34.99	9.14	48.83	**49.61**	37.99	41.40	40.87	27.53	23.71	43.16	12.07	56.84	**58.55**
Roystonea regia	33.33	0.00	**100.00**	0.00	0.00	**100.00**	0.00	**100.00**	**100.00**	25.00	0.00	**75.00**	0.00	0.00	25.00	0.00	50.00	50.00	28.57	0.00	**85.71**	0.00	0.00	40.00	0.00	66.67	66.67
Swietenia mahagoni	**100.00**	**100.00**	87.50	60.00	44.44	87.50	66.67	**100.00**	**100.00**	58.33	58.33	58.33	25.00	33.33	58.33	33.33	58.33	**66.67**	73.68	73.68	70.00	35.29	38.10	70.00	44.44	73.68	**80.00**
Eucalyptus citriodora	0.00	**0.00**	0.00	**100.00**	0.00	0.00	25.00	**100.00**	75.00	0.00	0.00	0.00	33.33	0.00	0.00	33.33	66.67	**100.00**	0.00	0.00	0.00	50.00	0.00	0.00	28.57	80.00	**85.71**
Mangifera indica	75.00	95.24	77.78	57.58	62.07	84.00	56.25	**96.43**	92.59	47.73	45.45	47.73	43.18	40.91	47.73	40.91	**61.36**	56.82	58.33	61.54	59.15	49.35	49.32	60.87	47.37	**75.00**	70.42
Tabebuia chrysantha	83.33	92.98	81.69	75.00	94.83	**94.92**	76.09	86.15	84.29	67.07	64.63	70.73	54.88	67.07	68.29	42.68	68.29	**71.95**	74.32	76.26	75.82	63.38	78.57	**79.43**	54.69	76.19	77.63
Delonix regia	23.81	80.00	50.00	12.50	36.36	41.67	0.00	64.29	**90.00**	38.46	30.77	30.77	7.69	30.77	38.46	0.00	**69.23**	**69.23**	29.41	44.44	38.10	9.52	33.33	40.00	0.00	66.67	**78.26**
Albizia falcataria	73.08	**93.75**	75.00	52.94	83.33	73.91	27.27	78.95	68.18	**59.38**	46.88	46.88	28.13	46.88	53.13	18.75	46.88	46.88	**65.52**	62.50	57.69	36.73	60.00	61.82	22.22	58.82	55.56
Ficus religosa	68.42	93.33	70.83	33.33	89.47	**93.75**	13.64	68.97	88.46	38.24	41.18	50.00	2.94	50.00	44.12	8.82	58.82	**67.65**	49.06	57.14	58.62	5.41	64.15	60.00	10.71	63.49	**76.67**
Terminalia arjuna	71.43	81.82	75.00	75.00	38.46	83.33	40.00	**85.71**	**85.71**	71.43	64.29	64.29	42.86	35.71	71.43	28.57	**85.71**	**85.71**	71.43	72.00	69.23	54.55	37.04	76.92	33.33	**85.71**	**85.71**
Bombax malabaricum	75.00	**100.00**	**100.00**	**100.00**	50.00	**100.00**	50.00	**100.00**	75.00	**30.00**	**30.00**	**30.00**	10.00	20.00	**30.00**	10.00	**30.00**	**30.00**	42.86	**46.15**	**46.15**	18.18	28.57	**46.15**	16.67	**46.15**	42.86
Archontophoenix alexandrae	**100.00**	**100.00**	**100.00**	50.00	40.00	**100.00**	22.22	85.71	**100.00**	55.56	55.56	55.56	11.11	22.22	55.56	22.22	**66.67**	**66.67**	71.43	71.43	71.43	18.18	28.57	71.43	22.22	75.00	**80.00**
Dimocarpus longan	76.92	90.00	84.62	58.33	76.92	**100.00**	75.00	**100.00**	68.75	52.63	47.37	57.89	36.84	52.63	47.37	47.37	**73.68**	57.89	62.50	62.07	68.75	45.16	62.50	64.29	58.06	**84.85**	62.86
Syzygium cumini	69.23	76.00	69.57	40.00	76.19	88.00	25.00	73.68	**91.18**	23.68	25.00	21.05	7.89	21.05	28.95	2.63	36.84	**40.79**	35.29	37.62	32.32	13.19	32.99	43.56	4.76	49.12	**56.36**
Plumeria rubra	71.43	75.00	44.44	50.00	83.33	**100.00**	18.75	81.82	88.89	38.46	46.15	30.77	15.38	38.46	38.46	23.08	**69.23**	61.54	50.00	57.14	36.36	23.53	52.63	55.56	20.69	**75.00**	72.73
Kigelia africana	0.00	0.00	0.00	0.00	0.00	33.33	0.00	0.00	**100.00**	0.00	0.00	0.00	0.00	0.00	**50.00**	0.00	0.00	**50.00**	0.00	0.00	0.00	0.00	0.00	40.00	0.00	0.00	**66.67**
Dracontomelon duperreanum	**100.00**	0.00	50.00	0.00	0.00	50.00	0.00	**100.00**	**100.00**	25.00	0.00	25.00	0.00	0.00	25.00	0.00	**50.00**	**50.00**	40.00	0.00	33.33	0.00	0.00	33.33	0.00	66.67	**66.67**
Ficus virens Aiton	53.33	87.50	71.43	57.14	80.00	53.85	40.00	76.92	**88.89**	36.36	31.82	22.73	18.18	36.36	31.82	18.18	**45.45**	36.36	43.24	46.67	34.48	27.59	50.00	40.00	25.00	**57.14**	51.61
Casuarina equisetifolia	60.00	**100.00**	83.33	62.50	66.67	**100.00**	50.00	75.00	88.89	66.67	66.67	55.56	55.56	66.67	66.67	55.56	66.67	**88.89**	63.16	80.00	66.67	58.82	66.67	80.00	52.63	70.59	**88.89**
Livistona chinensis	0.00	0.00	**100.00**	0.00	50.00	**100.00**	0.00	**100.00**	**100.00**	0.00	0.00	50.00	0.00	50.00	50.00	0.00	**100.00**	**100.00**	0.00	0.00	66.67	0.00	50.00	66.67	0.00	**100.00**	**100.00**
Ceiba insignis	42.11	46.15	53.33	37.50	66.67	35.29	19.05	65.00	**73.68**	32.00	24.00	32.00	24.00	24.00	24.00	16.00	52.00	**56.00**	36.36	31.58	40.00	29.27	35.29	28.57	17.39	57.78	**63.64**
Spathodea campanulata	85.71	91.18	88.24	77.78	93.33	96.67	77.27	97.14	**100.00**	75.00	77.50	75.00	52.50	70.00	72.50	42.50	85.00	**90.00**	80.00	83.78	81.08	62.69	80.00	82.86	54.84	90.67	**94.74**
unknown	87.50	**100.00**	**100.00**	91.30	86.96	**100.00**	91.30	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	95.24	**100.00**	**100.00**	**100.00**	**100.00**	93.33	**100.00**	**100.00**	95.45	90.91	**100.00**	95.45	**100.00**	**100.00**
Average	61.51	70.30	71.00	52.12	56.56	76.62	37.78	83.08	**87.20**	46.04	41.58	46.76	28.86	36.82	48.01	25.53	62.53	**65.60**	51.05	51.08	55.22	35.05	43.46	57.34	29.30	70.25	**73.47**



(a) (b) (c) (d) (e) (f) (g) (h) (i) (j)

Fig. 6. Visualization results of different comparison methods for individual tree segmentation and recognition in dense areas on the SZU-North hyperspectral dataset. The circled area indicates that the FBAN method significantly outperforms other comparison methods in segmenting and identifying individual trees in dense areas. The numerical representation of the model's predicted confidence that the region belongs to an individual tree. (a) Ground truth. (b) Mask R-CNN. (c) Cascade Mask R-CNN. (d) CondInst. (e) SOLOv2. (f) BoxInst. (g) Transfiner. (h) Maskformer. (i) Mask2former. (j) FBAN.



(a) (b) (c) (d) (e) (f) (g) (h) (i) (j)

Fig. 7. Visualization results of different comparison methods for individual tree segmentation and recognition in areas with different crown sizes on the SZU-North hyperspectral dataset. In the circled area, the FBAN method significantly outperforms other comparison methods in segmenting and identifying individual trees with different crown sizes. The numerical representation of the model's predicted confidence that the region belongs to an individual tree. (a) Ground truth. (b) Mask R-CNN. (c) Cascade Mask R-CNN. (d) CondInst. (e) SOLOv2. (f) BoxInst. (g) Transfiner. (h) Maskformer. (i) Mask2former. (j) FBAN.

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TABLE VII
SEGMENTATION AND RECOGNITION RESULTS OF SEVEN DIFFERENT COMPARISON METHODS ON SZU-SOUTH HYPERSPECTRAL DATASET

Method	AP	AP_{50}	AP_{75}	AP_s	AP_m	Confidence
Mask R-CNN	41.64	67.72	44.35	23.18	55.59	96.16
Cascade Mask R-CNN	43.80	71.18	47.32	24.24	57.98	94.84
Condinst	41.20	71.64	37.93	22.74	58.47	84.32
Solov2	33.46	57.49	36.45	20.17	43.93	91.56
BoxInst	19.64	59.10	10.95	14.50	23.30	86.97
Transfiner	29.95	69.15	19.52	16.22	44.23	97.27
Maskformer	11.46	18.84	11.88	6.37	16.85	87.56
Mask2former	49.82	74.21	58.21	34.25	68.66	97.04
FBAN	**54.32**	**78.36**	**62.69**	**39.59**	**72.03**	**98.02**

extent. Moreover, the SOLOv2 model performs poorly on the dense SZU-North dataset, with its metrics being only slightly higher than those of Maskformer. The visualization results in Figs. 6 and 7 clearly show that the segmentation recognition performance of the SOLOv2 model is similar to that of Maskformer. In complex scenes, the background and foreground may share many similar features, making it difficult for the Maskformer model to distinguish between the target and the background, leading to cases of mis-segmentation or missed segmentation.

The experimental results on the dense SZU-North dataset in Table V show that the segmentation and recognition indicators of the four comparison models Mask R-CNN, Cascade Mask R-CNN, CondInst, and Transfiner are similar. Based on Mask R-CNN, Cascade Mask R-CNN improves segmentation accuracy by cascading multiple detection heads. This implies that Cascade Mask R-CNN performs slightly better than Mask R-CNN in certain complex scenarios, but the overall metrics are not significantly different from those of Mask R-CNN. The CondInst model, which employs conditional convolutions, exhibits advantages in both flexibility and accuracy. Nevertheless, its segmentation recognition metrics on the dense SZU-North dataset are similar to those of Mask R-CNN and Cascade Mask R-CNN, showing relatively consistent results. The reason may be that their feature extraction strategies may not be sufficient to cope with the large amount of details and texture information in complex scenes. As shown in the first row of Fig. 6, all three models misidentified *Melaleuca leucadendra* Linn tree species as *Dracontomelon duperreanum* tree species. The *Plumeria rubra* tree species in the second row and the *Bauhinia purpurea* L. tree species in the third row were both missed. Especially in Fig. 7, the small-sized *E. decipiens* Hemsl. tree species is directly ignored. The performance of these four models in dense and overlapping tree areas also reveals their deficiencies in feature extraction, resulting in poor segmentation and recognition performance.

Transfiner achieves an AP of 30.86%, ranking moderately among various methods and significantly outperforming the aforementioned three approaches. However, compared to the top-performing FBAN, Transfiner still exhibits a notable performance gap. Specifically, its performance at higher IoU thresholds ($AP_{75} = 32.80\%$) is somewhat limited, and its effectiveness on small targets ($AP_s = 15.52\%$) shows clear limitations. Moreover, Transfiner performs poorly in certain specific classes. For instance, as shown in Table VI, both the recall and $F1$-score for *Eucalyptus citriodora* are 0, likely due to an insufficient number of samples for this category or challenges in capturing its distinctive shape features. A comprehensive analysis suggests that these shortcomings are primarily attributable to Transfiner's reliance on the Transformer's global feature modeling capability, which excels at capturing contextual information over a wide spatial range. However, this emphasis on global modeling may compromise the network's ability to capture fine-grained details, leading to suboptimal segmentation results in small or intricate regions, such as tree crown boundaries. This limitation is particularly evident in the missed detections of small target categories like *P. rubra* and *B. purpurea* L. in Fig. 6, as well as *Albizia falcataria* and *E. decipiens* Hemsl. in Fig. 7.

Although Mask2former employs a stronger decoder than Maskformer, leading to better segmentation recognition performance compared to other comparison methods, its performance is still slightly inferior to the proposed FBAN model. For example, the AP_{75} under the average precision at a high IoU threshold is still 4.41% lower than that of the FBAN model. The trees in the first row of Fig. 6 and the trees in the second row of Fig. 7 are misidentified. The main reason might be that both Mask2former and the aforementioned comparison methods are designed based on models for natural images, overlooking the extraction of spectral characteristics specific to different tree species. As shown in the red circle in the second row of Fig. 7, the segmentation effect is unsatisfactory in the area where multiple trees of the same species overlap. Additionally, Mask2former relies on the advantages of the Transformer architecture, which can effectively capture long-distance dependencies in images and excels in processing global features. However, it ignores the fine processing of local information, resulting in poor performance when segmenting objects in small scales and dense areas. Its AP_s for small-scale objects is only 20.03%. Small-sized tree species such as *Spathodea campanulata* and *B. purpurea* L. in the third row of Fig. 6 are also missed in segmentation. Also, *E. decipiens* Hemsl. in the second row of Fig. 7 is misidentified.

In the experimental results, the proposed FBAN model outperforms all other comparison models in segmentation recognition metrics. Specifically, the AP of the FBAN model is 44.08%, which is significantly higher than other compared models. This indicates that the FBAN model can achieve high segmentation accuracy across various scales and complex scenes. In addition, its confidence level is 95.57%, indicating that the reliability of its prediction results is very high. On the SZU-North dataset with densely overlapping trees and different scales in Figs. 6 and 7, the visual segmentation recognition results of the FBAN model are significantly better than other comparison methods, especially in areas that are difficult to process by other models. For example, in the second row of visualization results in Fig. 6, we can clearly see that in dense overlapping areas that are difficult to segment by all methods, the segmentation and recognition results of the FBAN model are more accurate and coherent, the segmented tree outlines are closer to the ground truth, and the details are more delicate. This further proves its excellent performance in processing complex scenes and diverse targets.

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(a) (b) (c) (d) (e) (f) (g) (h) (i) (j)

Fig. 8. Visualization results of different comparison methods for individual tree segmentation and recognition in dense areas on the SZU-South hyperspectral dataset. The circled area indicates that the FBAN method significantly outperforms other comparison methods in segmenting and identifying individual trees in dense areas. The numerical representation of the model's predicted confidence that the region belongs to an individual tree. (a) Ground truth. (b) Mask R-CNN. (c) Cascade Mask R-CNN. (d) CondInst. (e) SOLOv2. (f) BoxInst. (g) Transfiner. (h) Maskformer. (i) Mask2former. (j) FBAN.

TABLE VIII

SEGMENTATION AND RECOGNITION RESULTS OF DIFFERENT CLASSES BY DIFFERENT COMPARISON METHODS ON SZU-SOUTH HYPERSPECTRAL DATASET (BOLD NUMBERS INDICATE THE BEST CLASSIFICATION PERFORMANCE)

Class names	Precision									Recall									F1-score								
	Mask R-CNN	Cascade Mask R-CNN	Condinst	Solov2	BoxInst	Trans-finer	Mask-former	Mask2-former	FBAN	Mask R-CNN	Cascade Mask R-CNN	Condinst	Solov2	BoxInst	Trans-finer	Mask-former	Mask2-former	FBAN	Mask R-CNN	Cascade Mask R-CNN	Condinst	Solov2	BoxInst	Trans-finer	Mask-former	Mask2former	FBAN
Delonix regia	94.12	**100.00**	93.75	**100.00**	40.00	75.00	83.33	**100.00**	**100.00**	**94.12**	**94.12**	88.24	82.35	35.29	70.59	29.41	**94.12**	**94.12**	94.12	**96.97**	90.91	90.32	0.38	72.73	43.48	**96.97**	**96.97**
Artocarpus parvus	75.00	66.67	**87.50**	85.71	58.33	60.00	0.00	75.00	77.78	60.00	60.00	**70.00**	60.00	**70.00**	60.00	0.00	60.00	**70.00**	66.67	63.16	**77.78**	70.59	63.64	60.00	0.00	66.67	73.68
Wodyetia bifurcata	86.67	88.24	92.31	**92.86**	55.56	77.78	50.00	84.21	71.43	59.09	68.18	54.55	59.09	45.45	63.64	4.55	**72.73**	68.18	70.27	76.92	68.57	72.22	50.00	70.00	8.33	**78.05**	69.77
Plumeria rubra	0.00	33.33	33.33	0.00	0.00	**100.00**	0.00	66.67	**100.00**	0.00	33.33	33.33	0.00	0.00	66.67	0.00	66.67	**100.00**	0.00	33.33	33.33	0.00	0.00	80.00	0.00	66.67	**100.00**
Phoenix sylvestris Roxb	93.75	**100.00**	90.63	96.67	96.67	93.55	**100.00**	**100.00**	**100.00**	96.77	93.55	93.55	93.55	93.55	93.55	67.74	96.77	**100.00**	95.24	96.67	92.06	95.08	95.08	93.55	80.77	98.36	**100.00**
Koelreuteria paniculata	92.86	**100.00**	**100.00**	90.91	92.86	93.33	57.14	**100.00**	93.33	86.67	86.67	**93.33**	66.67	86.67	**93.33**	26.67	80.00	**93.33**	89.66	92.86	**96.55**	76.92	89.66	93.33	36.36	88.89	93.33
Choerospondias axillaris	69.70	60.98	**100.00**	18.75	63.41	68.85	22.22	68.09	69.44	23.47	25.51	5.10	3.06	26.53	**42.86**	2.04	32.65	25.51	35.11	35.97	9.71	5.26	37.41	**52.83**	3.74	44.14	37.31
Celtis sinensis	**100.00**	**100.00**	**100.00**	80.00	87.50	93.75	57.14	**100.00**	92.86	**100.00**	**100.00**	**100.00**	50.00	87.50	93.75	25.00	81.25	81.25	**100.00**	**100.00**	**100.00**	61.54	87.50	93.75	34.78	89.66	86.67
Senna siamea	64.66	80.95	86.42	71.64	77.78	87.06	61.90	**90.48**	86.05	71.43	64.76	66.67	45.71	60.00	70.48	24.76	**72.38**	70.48	67.87	71.96	75.27	55.81	67.74	77.89	35.37	**80.42**	77.49
Leucaena leucocephala	77.12	81.73	88.89	78.16	88.66	71.90	64.44	84.31	**89.47**	**79.13**	73.91	76.52	59.13	74.78	75.65	25.22	74.78	73.91	78.11	77.63	**82.24**	67.33	81.13	73.73	36.25	79.26	80.95
Cinnamomum camphora	72.73	**96.00**	85.71	60.42	82.35	83.87	90.91	84.00	80.30	49.38	59.26	59.26	35.80	51.85	64.20	12.35	51.85	**65.43**	58.82	**73.28**	70.07	44.96	63.64	72.73	21.74	64.12	72.11
Lagerstroemia indica	33.33	50.00	50.00	**100.00**	**100.00**	**100.00**	0.00	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	0.00	**100.00**	**100.00**	50.00	66.67	66.67	**100.00**	**100.00**	**100.00**	0.00	**100.00**	**100.00**
unknown	75.00	**100.00**	**100.00**	**100.00**	80.00	66.67	**100.00**	77.78	**100.00**	66.67	55.56	55.56	55.56	44.44	44.44	22.22	77.78	**88.89**	70.59	71.43	71.43	71.43	57.14	53.33	36.36	77.78	**94.12**
Average	71.92	81.38	85.27	75.01	71.01	82.44	52.85	86.96	**89.28**	68.21	70.37	68.93	54.69	59.70	72.24	18.46	73.92	**79.32**	67.42	73.60	71.89	62.42	63.88	76.45	25.94	79.31	**83.26**

2) Experimental Results on the SZU-South Dataset: On the SZU-South dataset, the Maskformer model exhibits poor segmentation performance, achieving an AP of only 11.46% with a confidence score of 87.56% (as shown in Table VII). This indicates that the overall segmentation performance of Maskformer on this dataset is not satisfactory. The visualization results of Maskformer in Fig. 8 show that the trees are barely segmented and recognized, indicating its limitations in dealing with fine-grained objects and complex backgrounds. The Maskformer framework is more suitable for processing global features, but may not perform as well as traditional convolutional networks (such as Mask R-CNN and SOLOv2) in processing boundaries and details. Although SOLOv2 performs slightly better than Maskformer on the SZU-South dataset, its segmentation metrics are still quite low. It can be observed in Fig. 8 that the detection boxes of individual trees are incorrectly detected and some small-sized trees are incorrectly identified. This may be due to the fact that background interference in complex scenes increases the mis-segmentation rate of SOLOv2, especially in natural environments where complex backgrounds such as land and vegetation have a greater impact on segmentation. On this dataset, the segmentation recognition performance of Mask R-CNN and Cascade Mask R-CNN are still comparable, especially in dense areas, they can effectively segment and identify trees. However, for small-sized trees, Mask R-CNN and Cascade Mask R-CNN still exhibit instances of missed segmentation or mis-segmentation.

BoxInst performs well on certain tree species that are easy to segment, such as *Phoenix sylvestris* Roxb and *Lagerstroemia indica* in Table VIII, achieving precision and recall rates close to 100%. This demonstrates its strong adaptability and segmentation capability when handling objects with clear boundaries and large areas. However, as shown in Table VII, its overall performance is limited, with an AP of 19.64%, ranking among the lowest of all methods, and AP_{75} of just 10.95%, significantly lower than others. This indicates its challenges in complex scenarios, such as those with significant background noise or intricate object boundaries. Furthermore, due to its weak ability to capture boundary details, its segmentation performance on small targets is suboptimal ($AP_s = 14.50\%$). Specifically, for the classes *P. rubra* and *Choerospondias axillaris*, BoxInst exhibits low recall and $F1$-scores. Additionally, as observed in Fig. 9, there are noticeable omissions in segmenting small tree crowns, further underscoring its inadequacy in handling small targets and complex scenes.

Transfiner performs significantly better than BoxInst on segmentation tasks involving medium-sized targets ($AP_m = 44.23\%$) and achieves higher confidence (confidence $= 97.27\%$) due to its powerful global feature modeling capabilities, indicating highly reliable segmentation results. However, the Transformer-based architecture of Transfiner tends to prioritize capturing global contextual information while somewhat neglecting local details. This limitation leads to subpar performance at high IoU thresholds ($AP_{75} = 19.52\%$) and in small object segmentation ($AP_s = 16.22\%$). As shown in the

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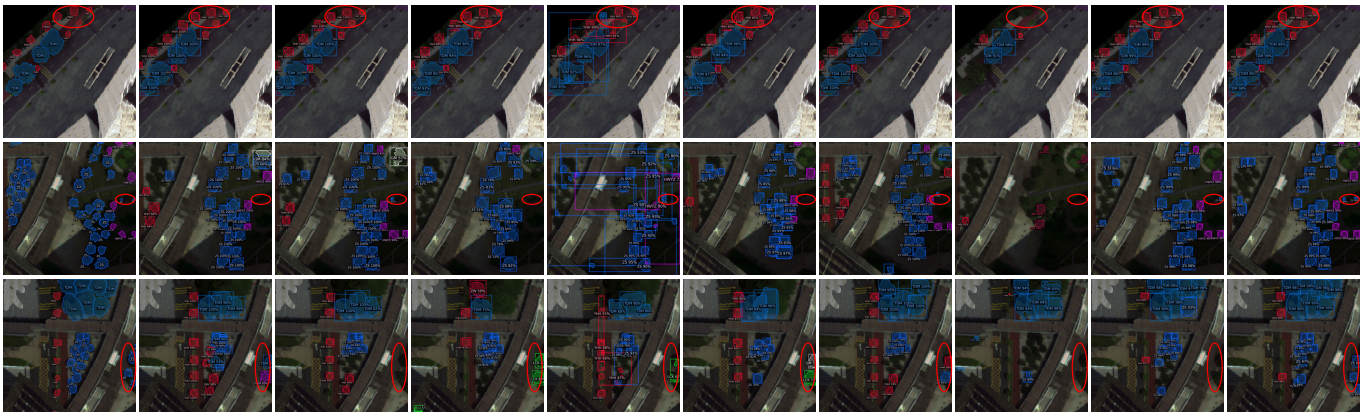


Fig. 9. Visualization results of different comparison methods for individual tree segmentation and recognition in areas with different crown sizes on the SZU-South hyperspectral dataset. In the circled area, the FBAN method significantly outperforms other comparison methods in segmenting and recognizing individual trees with different crown sizes. The numerical representation of the model's predicted confidence that the region belongs to an individual tree. (a) Ground truth. (b) Mask R-CNN. (c) Cascade Mask R-CNN. (d) CondInst. (e) SOLOv2. (f) BoxInst. (g) Transfiner. (h) Maskformer. (i) Mask2former. (j) FBAN.

third row of Fig. 8, Transfiner struggles to accurately identify smaller tree canopies at the bottom.

Even though Mask2former is an image segmentation model based on masking technology and self-attention mechanism, it can efficiently and accurately perform segmentation on few-sample and small-sample images. Nevertheless, trees usually have complex shapes and small branches, and these details are easily ignored or diluted during feature extraction, resulting in segmentation failure. The most intuitive manifestation is that in the red circles in the second and third rows of Fig. 9, all small-sized trees are missed. The FBAN model still performs best on this dataset, with an AP of 54.32%. The segmentation and recognition results shown in Fig. 8 further demonstrate the superiority of the FBAN model in dealing with densely overlapping trees. The reason is that the BAM of the FBAN model accurately captures boundary information and refines segmentation results through channel boundaries, spatial boundaries, and spectral boundaries. It effectively highlights important features and suppresses background noise, allowing it to accurately segment each tree in complex scenes, significantly outperforming other comparison models. Additionally, FBAN's segmentation metrics for trees of different scales are also significantly better than those of other models, especially with a confidence level reaching 98.02%. Its AP_s and AP_m are also 39.59% and 72.03%, respectively. These data indicate that FBAN performs well not only on small targets but also excels on medium and large targets. The visualization results in Fig. 9 also support this conclusion. This is due to the fact that ACAM in the FBAN model adaptively learns weights through convolution to adaptively represent canopy features of different sizes, significantly improving the flexibility and accuracy of the model.

E. Evaluation on Feature Backbone Robustness

Table IX presents the experimental results of the proposed FBAN framework's backbone feature extractor combined with the decoders of Maskformer and Mask2former. The table shows that the proposed FBAN framework's backbone feature extractor, when combined with Maskformer's simple decoder, still achieves optimal performance. Specifically, its precision reaches 71.19%, nearly double that of Maskformer. This indicates that despite the simplicity of the Maskformer decoder, the model still achieves excellent performance with the support of the powerful backbone feature extractor. Additionally, after combining with the Mask2former decoder, almost all indicators are significantly better than those of other models. For instance, its AP reaches the highest value of 44.08% among all models. This confirms that the backbone feature extractor of the proposed FBAN framework plays a vital role in the entire model, providing rich and accurate individual tree features, thereby enhancing overall segmentation and recognition performance. Moreover, at a 75% IoU threshold, the proposed FBAN framework achieves an AP_{75} score of 50.49. This demonstrates its superior performance under stricter matching conditions, maintaining excellent individual tree segmentation and recognition capabilities even when high precision is required. In summary, whether combined with a simple decoder or a complex one, the backbone feature extractor of the FBAN framework is crucial in enhancing model performance. It provides reliable and efficient solutions for various segmentation and recognition tasks in practical applications.

TABLE IX

EVALUATION AND ANALYSIS OF THE ROBUSTNESS OF THE BACKBONE FEATURE EXTRACTOR IN THE PROPOSED FBAN FRAMEWORK

Model	AP	AP_{50}	AP_{75}	AP_s	AP_m	Precision	Recall	F1-score
Maskformer	17.77	26.02	19.50	9.07	19.84	37.78	25.53	29.30
FBAN+Maskformer	28.03	43.46	31.33	11.25	30.69	71.19	43.91	52.86
Mask2former	40.57	61.24	46.08	20.03	44.70	83.08	62.53	70.25
FBAN+Mask2former	**44.08**	**65.29**	**50.49**	**22.01**	**47.98**	**87.20**	**65.60**	**73.47**

F. Model Performance Analysis

A comprehensive evaluation of various models' performance metrics, including feature activation maps, average time, and frame rate on the test set, significantly enhances

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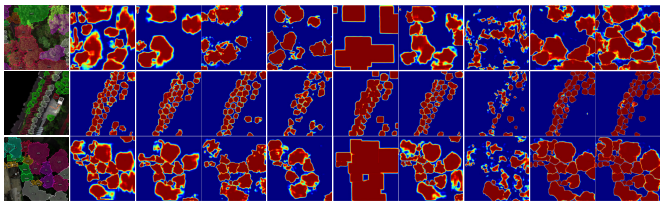


Fig. 10. Feature activation maps of different comparison methods on the SZU-North dataset. The color intensity indicates the probability that the pixel represents a tree. (a) Ground truth. (b) Mask R-CNN. (c) Cascade Mask R-CNN. (d) CondInst. (e) SOLOv2. (f) BoxInst. (g) Transfiner. (h) Maskformer. (i) Mask2former. (j) FBAN.

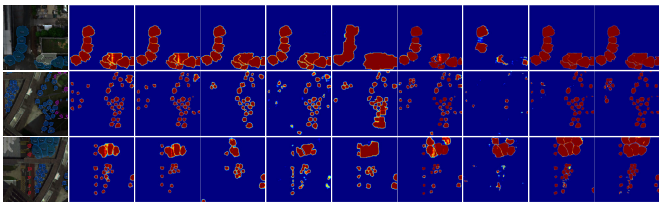


Fig. 11. Feature activation maps of different comparison methods on the SZU-South dataset. The color intensity indicates the probability that the pixel represents a tree. (a) Ground truth. (b) Mask R-CNN. (c) Cascade Mask R-CNN. (d) CondInst. (e) SOLOv2. (f) BoxInst. (g) Transfiner. (h) Maskformer. (i) Mask2former. (j) FBAN.

the understanding of their adaptability across different environmental contexts.

1) Feature Activation Mapping: In the tasks of tree species segmentation and recognition, tree crown feature activation maps can intuitively help understand the degree of attention the model pays to different areas of the image. Figs. 10 and 11 represent the feature activation maps of different comparison methods in the SZU-North and SZU-South datasets, respectively. The color intensity indicates the probability that each pixel belongs to a tree (values range from 0 to 1). In Fig. 10, the trees are densely packed, so it can be clearly seen that the comparison methods perform poorly in handling these complex overlapping structures, resulting in blurred or erroneous boundaries. In particular, the Maskformer method performs poorly in the precise extraction of tree crown boundaries. In particular, the Maskformer method performs poorly in accurately extracting the boundaries of tree crowns. Other methods may provide slight improvements, but there are still cases of missed segmentation and mis-segmentation. However, the proposed FBAN method has a BAM module designed to extract the crown boundaries of different trees from 3-D: channel, spatial, and spectral. Therefore, its crown boundary feature extraction effect in dense areas is better than other methods.

Furthermore, from Fig. 11, it is evident that the tree crown boundaries in (d) and (e) appear lighter in color. This suggests that the two comparison methods represented, CondInst and SOLOv2, are not optimal for accurately extracting the crown boundary features of individual trees. Mask R-CNN and Cascade Mask R-CNN show slightly better feature extraction performance, but their boundary determination in dense areas remains unstable. The only comparable method is the Mask2former method, but it can be clearly seen in (g) that many small tree crowns are missed. Instead, the proposed FBAN method stably extracts the boundary features of tree crowns of different sizes through the ACAM module, effectively coping with the diversity and complexity of trees in the forest.

2) Efficiency Analysis: During the testing experiments, a single 24GB RTX 3090 graphics card is being utilized, with a batch size set to 1. Fig. 12 records the average time and frames per second (FPS) of different models on the test set. The average time is the mean test time for all images, while the FPS is the number of image frames the model can process in one second. Therefore, the smaller the average time required by the model, the better, and the higher the FPS, the better. The figure offers an intuitive understanding that the FBAN model's average time and FPS are above average. Compared to other models, its performance is quite impressive, with both its average time and FPS approaching the best results. Overall, although the average time and FPS of the FBAN model are not the best among all comparison methods, it provides the most accurate and efficient detection performance. This makes it especially suitable for the task of individual tree species segmentation in dense areas, meeting various practical application needs.

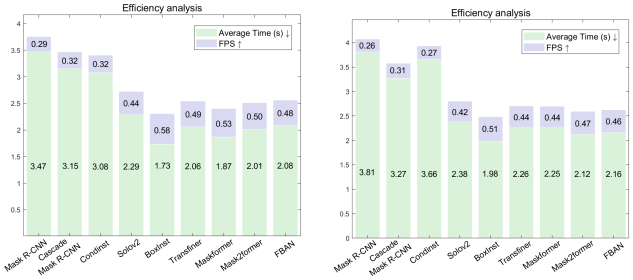


Fig. 12. Analysis of the average time and FPS for the FBAN model. (a) SZU-North dataset. (b) SZU-South dataset.

V. Conclusion

To address the challenge of accurately extracting the crown features of individual trees in dense forests, this article proposes the FBAN method for hyperspectral individual tree segmentation and recognition in dense forests. This method significantly enhances the ability to extract tree boundary features and improves the accuracy and stability of segmentation and recognition in dense forest areas. In the proposed FBAN, a BAM module is constructed to extract the boundary features between different trees, especially the connection boundary features between trees of the same species, through channel attention, spatial attention, and spectral attention. In addition, to further enhance the ability to extract crown features of different sizes, the ACAM module is designed. This module couples the self-attention mechanism and dilated convolution to adaptively fuse multiscale crown features, thereby improving the model's performance in complex scenes. Ablation experiments and comparative experiments on different types of hyperspectral datasets verify the accuracy and robustness of

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the proposed FBAN model for individual tree segmentation in dense and complex forest scenes. This method helps accurately understand the current status of forest resources, optimize forest management strategies, maximize forest carbon sequestration capacity, and achieve both ecological and economic benefits.

In a complex forest environment with a wide variety of tree species and dense crowns, the proposed FBAN model can solve the problems of dense distribution of similar tree species and multiscale crown segmentation to a certain extent. However, in extremely complex scenes, especially in areas with complex lighting conditions or severe occlusion, the segmentation and identification of individual trees still have certain limitations. Therefore, future research will focus on the fusion of multimodal models and explore methods to combine hyperspectral data with other sensor data (such as point cloud data and RGB data). By fully leveraging the advantages of different data sources, it is expected to further improve the accuracy and robustness of tree segmentation and recognition in complex forest scenes.

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